Intrahepatic Cholestasis of Pregnancy guideline (GL880)

Approval

Approval Group	Job Title, Chair of Committee	Date
Maternity & Children's Services Clinical Governance Committee	Chair Maternity Clinical Governance Committee	10th January 2025

Change History

Version	Date	Author, job title	Reason
10	April 2021	M Pereira, Consultant Obs & Gynae	Reviewed, minor changes and original appendix 2 removed as LFTs monitoring is now stored on EPR by DAU team (pg. 3)
11	November 2022	B Chohan, Consultant O&G S Geddes, Antenatal services deputy manager	Reviewed and updated to reflect current practice going forward
11.1	March 2023	S Geddes, Antenatal services manager	Live change to update flowchart in appendix 2
11.2	November 2024	B Chohan, Consultant O&G	Biennial review – no change

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1. Overview

Intrahepatic Cholestasis of Pregnancy (ICP) is a multifactorial condition of pregnancy characterised by an intense pruritus (itch), in the absence of a skin rash with abnormal maternal bile acid concentrations. The onset of symptoms is most common in the third trimester, but can be earlier in pregnancy. (1)

In the UK, ICP affects about seven in 1000 women (less than 1%). but increases to 1.2 -1.5% of Asian population. (1)

Genetic and environmental factors influence the prevalence worldwide.

2. Presenting symptoms and signs

Women with ICP present with

- Intense generalized itching classically affecting the palms and/or soles which is worse at night
- There is no associated rash
- Dark urine and/or pale stools (infrequent), steatorrhoea
- Jaundice is rare (<1%)

The pruritus is caused by excess bile acids in circulation, which stimulate nerve endings.

Most itching in pregnancy is not due to this disorder. Three out of four cases of itching in pregnancy are due to other benign causes.

Women with previous history of contraceptive pill induced cholestasis, or obstetric cholestasis in previous pregnancy or multiple pregnancy are at an increased risk.

3. Risks

3.1 *Maternal*

The incidence of pre-eclampsia is higher (12.2% compared to 3.4%), therefore it is important to assess blood pressure and urine alongside all ICP reviews.

There is a higher incidence of gestational diabetes, however additional screening is not recommended and should follow the national guidance for gestational diabetes screening.

There is an increased likelihood of being diagnosed with later hepatobiliary disease predominantly due to gallstones, however as gallstones are so common, causation is unclear.

There is no increased risk of postpartum haemorrhage.

Stillbirth remains the major concern. The risk of stillbirth appears to be correlated to the PEAK level of BAs (see below for further details on mild, moderate and severe ICP). For singleton pregnancies, the risk of stillbirth is only increased above a population rate once the BA levels are >100μmol/L. A recent individual patient data meta-analysis of 5269 women with OC demonstrated a clear threshold of serum bile acids >100μmol/L, above which the risk of stillbirth increased to 3.5%; there was no increased risk of stillbirth for women with bile acids below this level compared to matched international population data. The pathophysiology is uncertain but is suggested to be caused by BAs causing an acute fetal anoxic event perhaps due to fetal arrhythmia or acute placental vessel spasm.

Stillbirth with ICP in multifetal pregnancies is multifactorial, but stillbirth with ICP in twin pregnancies is higher and may occur sooner compared to singleton pregnancies.

There is an increased risk of both spontaneous and iatrogenic preterm birth with BA >40μmol/L.

There is an increased risk of meconium stained liquor, which is more likely to occur at lower gestational ages and more commonly between 35-38 weeks. Meconium is also more common with moderate to severe ICP.

There is a small increase in admission to the neonatal unit, with 30% being due to respiratory problems, and a median duration of stay of 7 days. [3]

4. Diagnosis & Antenatal Management

A history of itching, often but not exclusively on the palms and soles and worse at night, not associated with rash is suggestive of ICP. If the history is suggestive of ICP, send bloods for liver function tests (LFTs) and bile acids (BA). The BA should be a non-fasting sample and a level of >19μmol/L is taken as being diagnostic of ICP. If the BA is found to be raised, it should be rechecked a week later as mildly raised levels may normalise making the diagnosis of ICP obsolete.

Bile acid levels can go up and down. Subsequent frequency of biochemical assessment is as recommended below –

- Mild ICP (BA >19μmol/L to 39μmol/L): weekly testing from 38 weeks to aid discussion regarding timing of delivery
- Moderate ICP (BA >40μmol/L to 99μmol/L): weekly testing from 35 weeks to aid discussion regarding timing of delivery (in case levels rise >100μmol/L)
- Severe ICP (BA >100μmol/L): further routine testing may not impact on decision making and therefore not be routinely required.

If the BA and LFTs are within normal range but there is ongoing itching, both should be repeated at 1-2 week intervals.

Routine use of other investigations to exclude other causes for the clinical picture is no longer recommended unless there is uncertainty or atypical features such excessively elevated LFTs, rapidly progressive biochemical profile or early presentation in the first or second trimester. In such women, consideration should be made to refer to a hepatologist to discuss investigations and treatment. Testing for hepatitis C if the woman is found to have ICP is also no longer recommended.

Cohort studies have reported no cases of prolonged prothrombin time in women with uncomplicated ICP, therefore routine testing for coagulation abnormalities is not recommended for uncomplicated ICP. It can be considered on an individual basis.

No additional scans are required during pregnancy as it is not associated with IUGR and scans do not predict stillbirth. There is no evidence that fetal monitoring by either ultrasound or CTG for this condition is of any value particularly, as there is no evidence of placental insufficiency.

Women diagnosed with ICP with BA >40µmol/L should be referred to ANC to discuss further management including recommendations for timing of delivery; those patients with BA >100µmol/L should be referred to Miss Brooks' ANC specifically.

The woman should be advised to have weekly outpatient blood tests for checking LFTs from the time of diagnosis. The DAU team has created a patient list on EPR for obstetrics cholestasis patients. Once ICP is diagnosed, invite patient for DAU appointment for an 'ICP Start'. During this appointment patients should be seen by a member of the obstetric team and given a prescription for medication as outlined below if indicated. They will be given forms for blood tests to be done as an outpatient as well as written information on the condition. DAU staff will update the results and further management once a week and inform the patients over the phone. If anyone needs access to this list, they may contact DAU team.

If there are concerns with rising or un-resolving biochemical markers, &/or worsening symptoms, antenatal clinic appointment will be organised by the DAU staff.

Treatment for ICP is aimed at managing the symptoms associated with ICP and do not improve the BA levels or prevent stillbirth. Topical emollients including aqueous cream with or without menthol may relieve some discomfort from itching. Antihistamines such as chlorphenamine may provide sedative effects and may be useful at night. Other non-sedative antihistamines may be considered such as loratidine and cetirizine.

5. Ursodeoxycholic Acid

Evidence from RCTs shows there is no reduction in adverse perinatal outcomes including stillbirth or early spontaneous preterm birth before 34 weeks, although late spontaneous preterm birth before 37 weeks is reduced in women using ursodeoxycholic acid (UDCA) compared to women in the placebo group. In the largest trial, BA concentrations were found to be higher in the UDCA treatment group; ALT levels were found to be lower but the significance of this is uncertain. There is also only a minimal effect on reducing itching symptoms.

Routine use of UDCA to reduce adverse perinatal outcomes in women with ICP is therefore not recommended. The optimal starting and dosing regimen of UDCA remains unclear, but use o can be considered in women in whom the BA level is >40µmol/L from 34-36 weeks which may reduce the incidence of late spontaneous preterm birth.

Use of UDCA should only be prescribed after consultation with a senior clinician (middle grade or above) as it is not licensed for use in pregnancy. The dose is 12mg/kg/day.

If clotting is abnormal, vitamin K (Menadiol 10mg daily) should be prescribed. There is no evidence from RCTS for the routine use of Rifampicin.

Treatments to improve maternal itching are of limited benefit. Itching is unrelated to BA levels. Topical emollients can be considered such as aqueous cream (with or without menthol) to ameliorate skin symptoms. Antihistamines can also be considered including chlorphenamine, which has sedative-effects and may be useful at night, or other non-sedative antihistamines such as loratadine and cetirizine.

6. Timing and delivery

A large systematic review and individual patient data meta-analysis of women with ICP has reported the risks of stillbirth as

- 0.13% with peak BA levels <39µmol/L which is no higher than the background population risk.
- 0.28 for peak BA levels 40 to 99µmol/L, which again is no higher than overall background population risk, although the risk appears to increase around 38-39 weeks.
- 3.4% for peak BA levels >100µmol/L, which is higher than the background population rate, especially from 35-36 weeks.

Given the variable risk of stillbirth, the following is recommended in terms of timing of delivery

- Peak BA levels 19-39μmol/L: if there are no other risk factors, consider option of planned birth by 40 weeks.
- Peak BA levels 40 to 99μmol/L: if there are no other risk factors, consider option of planned birth at 38-39 weeks.
- Peak BA levels >100μmol/L: if no other risk factors, consider option of planned birth at 35-36 weeks. Where women have already passed this gestation, induction of labour should be arranged imminently.

In women with ICP and peak BA levels >40μmol/L, co-morbidities (such as pre-eclampsia, diabetes or multifetal pregnancy) may be associated with a greater risk of stillbirth. The presence of additional risk factors needs to be taken into account when considering the timing of planned delivery.

In terms of monitoring in labour, CEFM should be offered to all women with BA levels >100μmol/L. whilst there is insufficient evidence relating to CEFM where BA levels are below this threshold, the recommendation for such patients is as follows –

- Peak BA levels 19-39μmol/L: provided there is no other indication for CEFM, these patients can have intermittent auscultation therefore homebirth and delivery on the birth centre are not contraindicated.
- Peak BA levels >40μmol/L: CEFM is recommended, as there is a higher likelihood of meconium and the baby needing neonatal care.

The presence of additional risk factors will likely necessitate the need for CEFM anyway, such as pre-eclampsia, diabetes etc. Women with moderate to severe ICP are more likely to experience the passage of meconium during labour, which should be, managed as per usual practice. Women should also be informed that with moderate to severe ICP, their baby is more likely to receive neonatal care.

Continuous electronic fetal monitoring (CEFM) is recommended for all women with BA ≥40 μmol/L on delivery suite.

7. Postnatal

Postnatal resolution of the condition should be confirmed four weeks after delivery by repeating BA and LFTs, which should be arranged by the GP. The levels should be checked sooner if the patient is unwell or if an additional diagnosis is suspected.

Women can choose to use any contraception of their choice, including the combined oral contraceptive pill (COCP) unless the woman has a history of combined hormonal contraception (oestrogen containing) related cholestasis who should then be advised to use progesterone-only or non-hormonal methods.

Women should also be informed of the high risk of recurrence in subsequent pregnancies. In subsequent pregnancies, baseline LFT and BA should be checked in order to establish they are normal. Levels should only be repeated if clinically indicated.

8. References

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Appendix 1 – Table of normal reference ranges and their interpretation in pregnancy

Table 1

Normal reference ranges and their interpretation during pregnancy *

Analytes	Normal (non-pregnant)	Pregnancy	Abnormalities and possible interpretations
Haemoglobin (g/dL)	11.5–16.5	11.0–15.0	Abnormal results need to be considered in conjunction with the patient's clinical state
White cell count (x 10^6 per mL)	4.0–11.0	Unchanged	
Platelets (x 10^6 per mL)	150–450	Unchanged	
Sodium (mmol/L)	135–145	132–140	Abnormal results need to be considered in conjunction with the patient's clinical state
Potassium (mmol/L)	3.5–5.5	3.2–4.6	
Urea (mmol/L)	2.5–6.8	1.0–3.8	↑ in: dehydration hyperemesis gravidarum late stages of pre-eclampsia renal impairment
Creatinine (mmol/L)	0.06–0.1	0.04–0.08	↑ in: renal impairment late stages of pre-eclampsia
Fasting glucose (mmol/L)	3.0–5.4	3.0–5.0	↑ in: gestational diabetes mellitus (refer to reference 3 for diagnostic criteria)
Total calcium (mmol/L)	2.2–2.60	2.0–2.40	↑ in: primary hyperparathyroidism
Ionized calcium (mmol/L)	1.16–1.30	1.16–1.30	
Magnesium (mmol/L)	0.6–1.0	0.6–0.8	↓ in: vomiting hyperemesis gravidarum
Albumin (g/L)	33–41	24–31	↓ in: malnutrition recurrent vomiting hyperemesis gravidarum
Bilirubin (micromol/L)	3–22	3–14	↑ in: intrahepatic cholestasis of pregnancy HELLP late stages of pre-eclampsia acute fatty liver viral hepatitides
Alanine aminotransferase (U/L)	1–40	1–30	↑ in: intrahepatic cholestasis of pregnancy HELLP late stages of pre-eclampsia acute fatty liver viral hepatitides
Aspartate aminotransferase (U/L)	1–30	1–21	↑ in: intrahepatic cholestasis of pregnancy HELLP late stages of pre-eclampsia acute fatty liver viral hepatitides
Alkaline phosphatase (U/L)	25–100	125–250	↑ in: metabolic bone disorders but placental serum alkaline phosphatase needs to be excluded

Adapted from reference 7

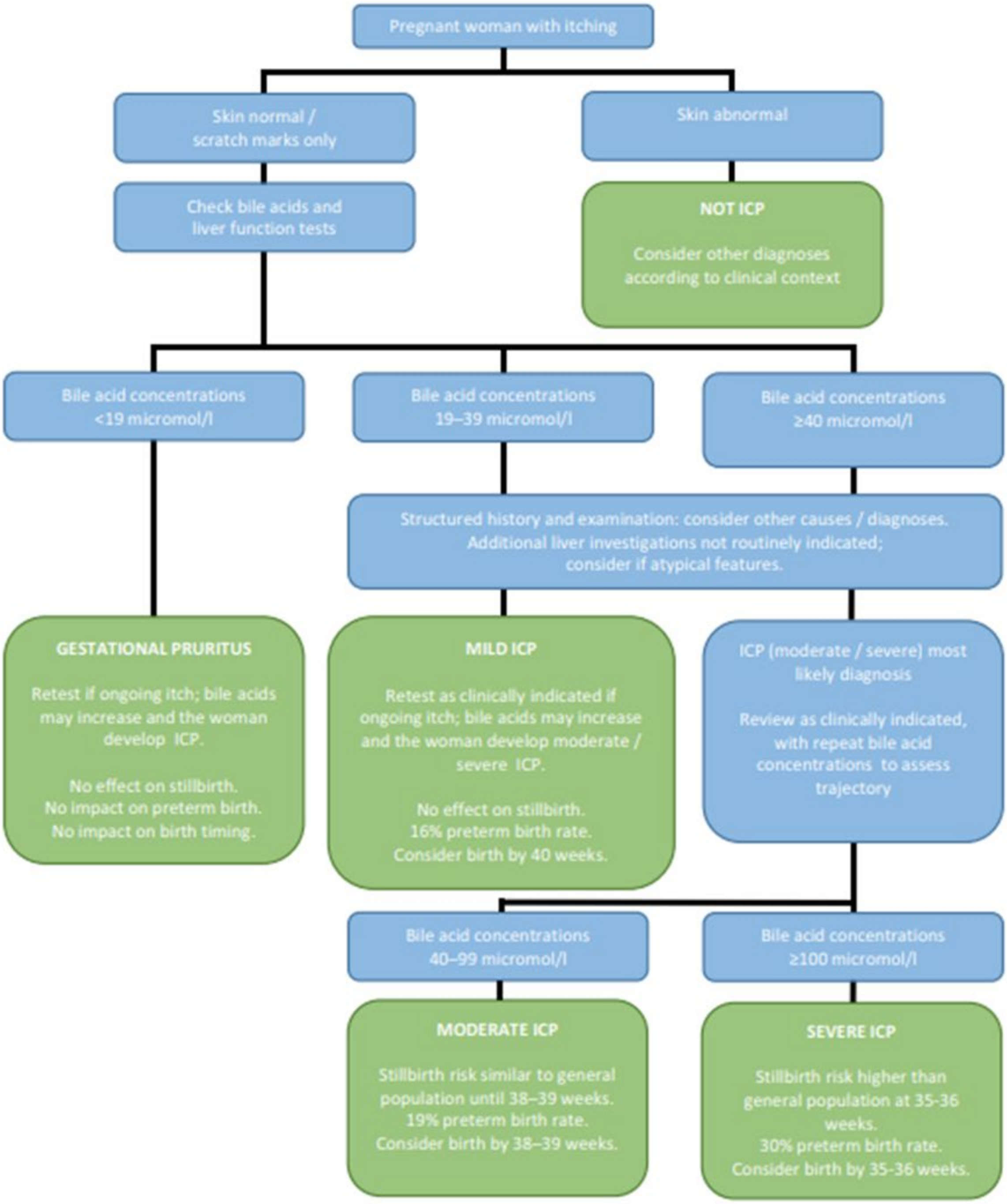
↑ increased concentration

↓ decreased concentration

HELLP Haemolysis-Elevated Liver enzymes-Low Platelets

* Each laboratory, where practicable, should develop its own reference ranges for pregnant women. Care should be exercised in comparing results from different laboratories due to differences in assay methodologies.

Appendix 2 – RCOG Flowchart for the care of pregnant women with itching (9)



Figures above relate to singleton pregnancy with no other risk factors.

Comorbidities (particularly pre-eclampsia and diabetes) or other obstetric risk factors (such as multifetal pregnancy), are associated with increased risk of stillbirth and should be taken into consideration when planning management.

Additional liver investigations may be considered in women with atypical features (e.g. early onset, marked transaminitis, jaundice, fever, or in whom postpartum resolution does not occur). These investigations may include liver ultrasound, viral hepatitis screen, liver autoimmune tests, and/or coagulation screen.